

# Supporting Circularity Flows through Temporal Change Analysis of Building Stock

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## Abstract

Digital Twins (DT) of cities offer potential for driving circularity in built infrastructure through cadastres of Material Stock (MS) of the city made available to the urban mining project. What was previously a static repository of MS and geo-reference data is now being made dynamic through the integration of Material Flow (MF) analysis that is constantly in development. While there has been much development in the capture of MS data, the temporal data of when the recycled material (RC) or recyclable material will be available is sparse. Demolition prediction models that are developed on visual inspection have a high accuracy, but visual cues may not show up till the building reaches vacancy or is close to demolition. The shortage of prior information about recoverable MS is a large deterrent for including RC in new construction design and planning. It also affects two cascading complications. The establishment of industry and processing units for recyclable material along with the development of appropriate technology requires pre planning and information on future availability of MS. Material storage between extraction and reuse is seldom possible for large MS in space critical urban development which also affects RC processes. As building stocks go through more complex processes of renovations and heritage conservation, a Weibull distribution would be insufficient to predict future demolitions. With this challenge in scope, a simple machine learning model is developed to make baseline predictions on building demolitions. The data frame which is built on the base MS data of (IOER) for Germany, uses open data sources for the sample study of Berlin and also gives insights to correlations that other parameters of built stock have with demolition probability.

## Background

Circularity in building material is largely dependent on accurate knowledge of Material Stocks and of the schedule of their availability (Uhlir 2023). This is done through a combination of top down (GIS and satellite imagery) and bottom up (Building level surveys, Scans etc) methods (Rajaratnam et al. 2025). The collection of MS has been explored through various macro and micro methods, a comprehensive modelling method for Germany was presented by Hartmann et al. 2016 using ATKIS (Hartmann et al. 2016). In 2025 the Leibnitz Institute for Ecological Spatial Development (IOER) published a Material Cadastre (Schinke et al. 2025) of the built stock in Germany. This is used by this paper and the analysis model and developed further. The cadastre Matcad2022 is developed through a bottom up approach (Ortlepp, Gruhler, and Schiller 2016) which classified 139 building types which are further grouped to 11 material classifier indicators based on 44 material. While this model does give a fair indication of material stocks at a national level, as the documentation itself suggests “*due to the type-based approach— the information provided on the individual buildings differs significantly from the materials actually used in the building. Targeted data aggregation of the building-related dataset is essential for a meaningful and appropriate interpretation of the data.*”- (Schinke et al. 2025).

The data uses an LOD2 (ibid.) level model from the 3D Building Metrics Germany 2022 which is also used as the base of this data frame. Although it is an LOD2 model, it works on an extrusion

of the building polygon (Ekanayaka, Popelka, and Burian 2026) and doesn't have critical data like upper floor offsets. This creates unreasonably large MS errors in podium type building morphology that makes up a large part of the urban building stock. Eg. the two Schlangenhader Tunnel buildings and the Rathaus Passagen building show this anomaly due to building morphology and show up as an MS that is several multiples higher than the onsite built mass.

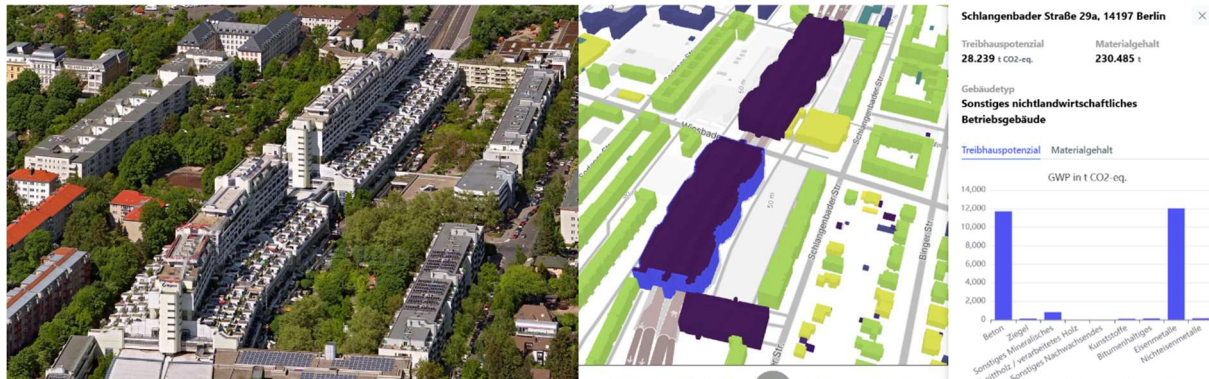


Fig 1-The image to the left shows the actual morphology of the podium type building and to the right is the extruded polygon model that misses this morphology to create an unreasonably large building mass and MS  
Image source: left- Nürnbergluftbild (Webref1), right- Concular website that makes the IOER data accessible (Webref2)

This presents two challenges when implied circularity is to be calculated - one, the building specific volume metrics, and two, building specific material metrics which still need to be modelled into DTs. But more granular data requires larger resources to be deployed. While access to LIDAR and point cloud technologies is fast increasing (Hartmann et al. 2016; Honic et al. 2023), and systemized bottom up processes are being enabled (Heeren and Hellweg 2019; Honic et al. 2023) which support the creation of a BIM to City Information Model integration (Brezina, 2023), a structured prioritized approach and probability planning to such documentation is necessary (Heeren and Hellweg 2019). A single building scan of a residential building could take up to 1.5 hrs (Honic et al. 2023). Apart from the infrastructure to record the infrastructure, to make such large polygonal data accessible through public servers is resource intensive. While material cadastres help in Pre Demolition Audits (Schiller et al. 2025) the gap between their granularity and that of visual verification models that verify through street view imagery or laser scans or integrate BIMs is large. As such both serve different purposes till they are fitted together through technology and digital infrastructure advancements. For the purpose of circularity, the collation of this data must be executed on the basis of a schedule that prioritises buildings that are threatened by demolition. In the recycling industry material flow it is a spatio-temporal process.

The profiling of spatiotemporal correlation of the built stock is critical for circularity (Kretschmar and Schiller 2023) which includes an understanding of how parameters like building types, location, density and energy changes in time affect the built mass. The understanding of this temporality is critical for development of industry and handling facilities that will be required in the future availability of certain recoverable building stocks. For eg. in Czech Republic where large amounts of plasterboard is being used now, the absence of any handling facilities to handle this material in the future when demolition mass of the material escalates and is to be addressed (Schiller et al. 2025) with spatiotemporal knowledge. The storage of Recovered building material in spatially constrained urban environments is one of the key challenges that is also a temporal event between deconstruction (Fehse, Franßen, and Braus 2025) and reuse or recycling which indicates the time pressure that circularity demands.

Given the above background, a baseline model is developed to improvise on the Matcad2022 (Schinke et al. 2025) data to preliminarily predict demolition. This model also analyses and presents the dataframe for other parameters that affect demolition probability.

## Methodology

Th Matcad2022 is used as the base layer, and subsequent layers are spatially joined to the base layer after cleaning of the data. Several data sources were explored, some were abandoned for unsuitable data fields or methods. Eg. the Berlin Energy Map (Nytsch, 2025) data was initially identified as a building level data source. But after EDA, it was found that the data was derived from Energy Indicators that were similar to the 11 Material Indicators of the base Matcad2022 model creating a more than 90% collinearity with it, as a result this was abandoned in favour of the block level energy data from Berlins geoportal (GDI, 2022). The primary data sources are given in the flow chart below.

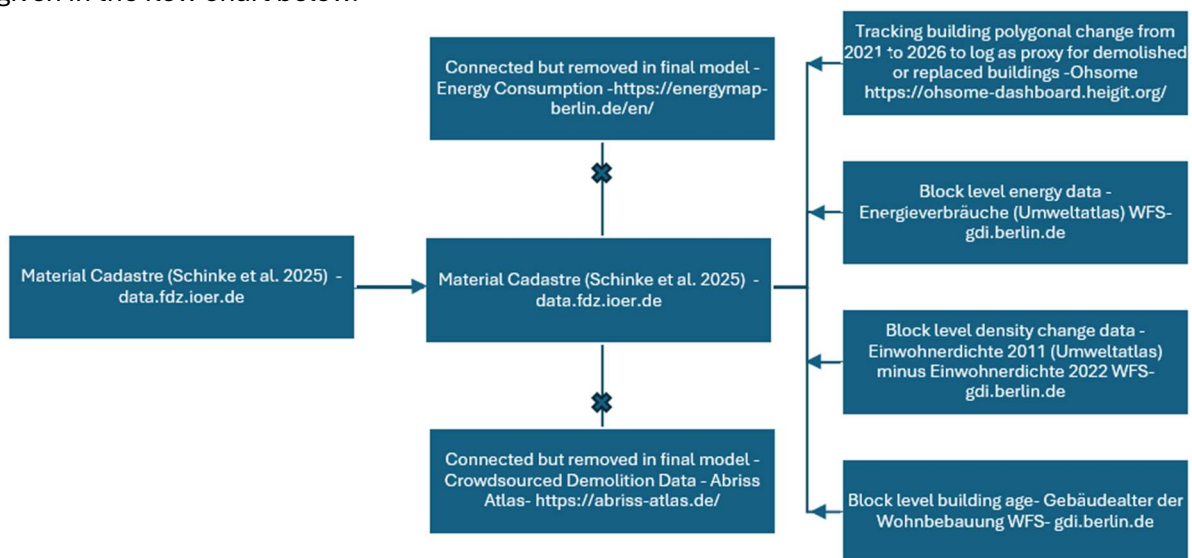


Fig 2-Showing the spatial join of all the layers and their data sources

Due to privacy concerns, a large amount of open data is only available in aggregated block level. Although the data holds individual counts, the assignment to spatial polygons has been removed to maintain privacy. This means that for data like building age, the assignment of a value is more to do with the distribution and spread of age of all the buildings in the block rather than the specific building age, but this is used as a parameter to signify how many new vs old buildings are present in the block etc. Similarly, the other parameters are drawn from aggregated counts.

Following the cleaning and checking of geometric alignment of the polygons, a spatial join is carried out. The final data used for analysis and training was a lean dataframe without any principal component analysis so that explainability is maintained. The extraction and the final models are shared at ....Ten demolition candidates were manually checked and found positive. The outliers of building volume were manually checked and found to be containing non retrievable anomalies and were dropped. The exploratory data analysis and the predictive model results will be discussed in the next section and collated with other research findings by previous authors.

## Analysis

The dataset is a more of a representative of the parameters given that it is drawn from aggregated data and not precise building level data, nonetheless it provides some critical information in modelling and understand of demolition probability detection. It is important to note that the demolition probability candidates were divided into two distinct groups - one where the building is replaced by another construction within the observation period i.e. 2021 to 2026, and another one where the site was left vacant after demolition. The two sets show distinguishable correlation with the other parameters of the data. 2021 was chosen as the beginning of the observation to buffer the base Matcad2022 dataset observations as the exact period of Matcad 3D observations or OSM observations could not be ascertained. The demolition binary is labelled as Geometry Change for buildings that have been replaced, Disappearance for buildings which were not replaced but the sites remained vacant after demolition, and Persistent for buildings that were not demolished or changed in the observation period. Given that the demolition candidature for the observation period of 2021/22-2026 was very small i.e. only 1730 buildings from 3,27,295 buildings, which is 0.53% or only one in every 189 buildings, the predictive model performance was expectedly weak. The data drawn through our method and official statistics align to the official records for 2022-2024 where 1412 buildings were demolished according to the Office of Statistics Berlin Brandenburg. The data for 2025 & 2026 was not yet released (d.o.r 16/09/2026).

## Building type association

Non-residential buildings show significantly higher rate of churn, i.e. higher demolition probability than residential buildings of which the multi-family house (MFH) is the most stable. This finding statistically resonates with the previous research on Non domestic buildings showing higher demolition rates (Kretzschmar and Schiller 2023) in our dataset. This pattern was also observed in crowd sourced data as the Abriss Atlas but was disregarded on the basis of observability where larger non-residential buildings are more visible than private houses. But through statistical evidence of our data this can now be confirmed. Keeping MFH as the baseline, storage buildings show 45 times the temporal rate percentage, followed by factories and trade buildings at 22x, the offices at 17x and so on. Interestingly single family homes show a 3 times higher demolition candidature rate than multi-family homes.

|                            | BtySig | STG      | PRO      | TRD      | OFC      | NAS      | HOT      | ONR      | HSN      | ASB      | SFH      | MFH      |
|----------------------------|--------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| total_classified           |        | 4160     | 7115     | 4033     | 4259     | 10348    | 1205     | 7652     | 1221     | 137      | 167994   | 95942    |
| disappearance_candidates   |        | 199      | 158      | 82       | 64       | 118      | 14       | 67       | 9        | 1        | 220      | 61       |
| geometry_change_candidates |        | 38       | 41       | 30       | 28       | 73       | 6        | 34       | 2        | 0        | 426      | 59       |
| temporal_change_candidates |        | 237      | 199      | 112      | 92       | 191      | 20       | 101      | 11       | 1        | 646      | 120      |
| disappearance_rate_pct     |        | 4.783654 | 2.220661 | 2.033226 | 1.5027   | 1.140317 | 1.161826 | 0.875588 | 0.737101 | 0.729927 | 0.130957 | 0.06358  |
| geometry_change_rate_pct   |        | 0.913462 | 0.576247 | 0.743863 | 0.657431 | 0.70545  | 0.497925 | 0.444328 | 0.1638   | 0.0      | 0.25358  | 0.061495 |
| temporal_change_rate_pct   |        | 5.697115 | 2.796908 | 2.777089 | 2.160131 | 1.845767 | 1.659751 | 1.319916 | 0.900901 | 0.729927 | 0.384538 | 0.125076 |

Fig. 3- the building type association with demolition candidature rates.

## Population Density Increase Correlation

It was observed that geometry change candidates lie in context with a high increase in population density, indicating the urgent need for rebuilding in these areas. Due to a low confidence index it could not be conclusively drawn that the disappearance candidates lie within the areas where the population density is lower.

|   | comparison                   | difference_median | ci_lower  | ci_upper |
|---|------------------------------|-------------------|-----------|----------|
| 0 | Disappearance - Persistent   | -0.099742         | -1.214291 | 1.403220 |
| 1 | Geometry-change - Persistent | 0.936525          | 0.262022  | 2.775268 |

Fig. 4 - shows the difference in density pressure between persistent and demotion candidates.

## Energy correlation

It was observed that the buildings in the replacement candidature of geometry change had a higher median energy consumption by 1.45kWh/m<sup>3</sup>, but demolition left vacant candidates show no significant deviation from mean energy consumption than non demolished buildings. This could be on account of vacancy but needs to be investigated further. The replaced buildings also show a higher energy consumption than the disappearance candidates.

|   | comparison                      | observed_difference | ci_lower  | ci_upper |
|---|---------------------------------|---------------------|-----------|----------|
| 0 | Disappearance – Persistent      | -0.326310           | -1.002751 | 0.288727 |
| 1 | Geometry-change – Persistent    | 1.451738            | 0.646069  | 2.257474 |
| 2 | Geometry-change – Disappearance | 1.778048            | 0.658696  | 2.856239 |

Fig. 5- The aggregated energy consumption correlation with demolition candidature.

At the building type level the Disappearance candidates show a consistently higher energy demand than median in single family houses, this is also observed for factory building in the geometry change candidature.

|    | BtySig | comparison                   | candidate_n | reference_n | median_difference | ci_lower  | ci_upper  | significant_95pct |
|----|--------|------------------------------|-------------|-------------|-------------------|-----------|-----------|-------------------|
| 10 | SFH    | Disappearance – Persistent   | 212         | 165748      | 3.085993          | 1.956822  | 3.851505  | True              |
| 6  | ONR    | Disappearance – Persistent   | 67          | 7471        | 1.582383          | -4.529940 | 11.356855 | False             |
| 14 | TRD    | Disappearance – Persistent   | 80          | 3886        | 0.760725          | -0.311612 | 1.855983  | False             |
| 4  | OFC    | Disappearance – Persistent   | 64          | 4123        | -1.046160         | -2.363949 | 1.108899  | False             |
| 7  | PRO    | Geometry-change – Persistent | 41          | 6856        | 3.809103          | 0.432448  | 8.220231  | True              |
| 0  | MFH    | Geometry-change – Persistent | 57          | 94861       | 0.909034          | -0.202099 | 3.472393  | False             |

Fig. 6- The aggregated energy consumption correlation with specific building types in the demolition candidature.

## Influence of age of surrounding building stock

As only anonymized age of every block was made available, the analysis followed secondary observations as to spread of building of various age groups in the block-age\_heterogeneity\_index and number\_of\_age\_periods to draw a correlation. This showed two distinct results, the disappearance candidates occur in contexts with higher than median age groups and also where the distribution of building of varied age groups is lower. More distinctly the change in geometry candidates are in context where the age is widely distributed showing already changing urban fabric.

When collated with other readings it can be drawn that where the population density pressure is higher there is already a changing age, and the older buildings in such contexts are more likely demolition candidates. On the other hand, where this pressure is not significant the context remains occupied by older buildings, and demolitions are not always followed by new buildings.

|   | comparison                                        | metric                  | candidate_n | reference_n | median_difference | ci_lower  | ci_upper  | significant_95pct |
|---|---------------------------------------------------|-------------------------|-------------|-------------|-------------------|-----------|-----------|-------------------|
| 0 | Disappearance/no-match candidate – Persistent ... | number_of_age_periods   | 976         | 299386      | -3.000000         | -3.000000 | -3.000000 | True              |
| 1 | Disappearance/no-match candidate – Persistent ... | dominant_age_share      | 583         | 257980      | -0.032432         | -0.038961 | -0.023342 | True              |
| 2 | Disappearance/no-match candidate – Persistent ... | age_heterogeneity_index | 583         | 257980      | -0.126316         | -0.462500 | 0.282353  | False             |
| 3 | Geometry-change candidate – Persistent classified | number_of_age_periods   | 713         | 299386      | 0.000000          | 0.000000  | 1.000000  | False             |
| 4 | Geometry-change candidate – Persistent classified | dominant_age_share      | 590         | 257980      | -0.055743         | -0.071146 | -0.040313 | True              |
| 5 | Geometry-change candidate – Persistent classified | age_heterogeneity_index | 590         | 257980      | 0.670517          | 0.310000  | 0.897512  | True              |

Fig. 7 -The contextual age of surrounding units with the demolition candidature.

## Predictive model

The predictive models were built without a principal component analysis and with a modest predictor dimension of six cleaned columns for explainability. Hence the predictions do not qualify as conclusive evidence, they rather lay the ground for further investigations based on statistical evidence. Of the Logistic Regression, Random Forest and Gradient Boosting models tested, the Random Forest model performed better than the others showing a ROC-AUC of 0.793 and Average precision of 0.0966 which may seem insignificant but is still relevant in a dataset with only 0.53% targets exist, which demonstrates that non-linear relationships exist between the predictors and the target. The enrichment shows the probability of the model in predicting a valid candidate for further inspection as compared to random selection. At 95.9 times higher precision, it could serve as valid first proofing tool. A more robust model including the 39 original columns with PCA could lead to even better results.

|   | Model                | Selected buildings | Actual candidates selected | Precision | Recall  | Random baseline | Enrichment over random |
|---|----------------------|--------------------|----------------------------|-----------|---------|-----------------|------------------------|
| 0 | Logistic Regression  | 7933               | 117                        | 1.475%    | 53.425% | 0.471%          | 3.1×                   |
| 1 | Random Forest        | 31                 | 14                         | 45.161%   | 6.393%  | 0.471%          | 95.9×                  |
| 2 | HistGradientBoosting | 8076               | 130                        | 1.610%    | 59.361% | 0.471%          | 3.4×                   |

Fig. 8 -Performance of the predictive model.

## **Discussion**

The data and the model present a simple machine learning tool collated from open data that can help assist in informing future demolition candidature and correlation with other building features such as energy consumption, age distribution, population density pressure and building type. It also identifies two separate candidates that inform us of possible vacancy post demolition, which could serve as sites of material storage which is an important challenge in circularity (Fehse et al. 2025). Its narrowing down of candidature pool could help prioritise buildings that require more precise volumetric and material modelling that is too resource intensive for capturing such data for the entire city's built stock.

More importantly for the recycling industry it is not a specific building that is important but capturing of certain trends like the demolition of warehouses in particular periods that will provide critical information on development of technologies to handle a particular waste classification. In Germany where about 90% construction waste is recovered but a large portion of it goes as landfill in reclamation measures (Ortlepp et al. 2016) development of a more robust predictive model will help develop industry around the prediction. Lastly as Uhlig (2023) has shown that recycled construction material is at times cheaper than equivalent primary materials, but is handicapped by other factors of which prior knowledge of material availability is one. With this knowledge, circularity can be better integrated into designs right from the design stage.

## **Shortcomings and Suggestions**

The data has no qualitative aspects captured which means it does not do justice to the present technologies of Artificial Intelligence available to handle such data. Further work involving qualitative aspects could not only make the data robust but also help us ask the right questions. This model was built to show an example of what is possible with aggregated data. In cases where precise building level data is available as recorded data not as estimates, the outcome could be much better. Also with qualitative inclusions the use of Neural Networks could be justified which was not employed for such a lean dataset. After the first layer of predictions, the use of computer vision models is suggested for proofing of the narrowed candidate pool and for collection of precise building façade indicators of material compositions. The use of YOLOv8 and Urban AI have shown promising results in diverse scenarios without requiring to manually label localized training data results (Raghu, Armeni, and De Wolf 2026; Raghu, Bucher, and De Wolf 2023)

## **Declaration on use of AI**

A strict qualitative vs quantitative approach is followed in AI use. Open AI support was used for programming in Jupyter Lab as a non integrated module but absolutely no part of the report text has been created using any AI tool whatsoever. The process of looking for relevant research papers, datasets and ideation of research was done without the support of any AI. Hence AI aided quantitative work but was kept out of any qualitative aspects that would affect the direction, outcome or analysis.

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